

Result-driven software engineer with a robust theoretical foundation and a passion for leveraging technology to tackle intricate challenges. Possesses strong communication, leadership, and collaborative skills, thriving in innovative team environments. Known for consistently meeting deadlines and delivering high-quality work under pressure. Eager to apply academic knowledge and practical skills to drive impactful contributions in technology driven organizations.



Biographic Information

Full Names:	Siphesihle Nqubeko
Surname:	Khumalo
Gender:	Male
Ethnicity:	African
Residential Address:	R252 Spoon Avenue Umlazi Durban
Postal Code:	4066
Availability:	Full-Time

Education

High School: Langazela Senior Secondary School
01/2014 – 12/2016

- Highest grade passed: Grade 12
- Stream: Science and Mathematics Stream

Tertiary: Tshwane University of Technology
01/2021 – 06sss/2024

- Qualification: Dip Computer Science
- Credits acquired: 360 Credits

- (Major Field)
- Software programming
 - Database design
 - System Analysis
 - Mobile device programming

Certificates & Rewards

- MAISH Hackathon participation
- Ayoba Business Chat API Hackathon participation
- Telkom Hackathon participation
- Operating Systems Tutor 2024
- Student Representative Council (Education and Transformation Office)
- Top Leadership Development Programs TUT
- CISCO badges (JavaScript essentials, Cybersecurity)

Interests

- Artificial Intelligence
- Data Science
- Playing Chess
- Blockchain Dev
- Economics
- Cloud Computing

Interpersonal Skills



Technical Skills

Programming Languages and Frameworks

- ✓ Java (JEE, JSE, JME & Springboot)
- ✓ Python (Django)
- ✓ JavaScript (React)
- ✓ Blockchain Smart Contracts (Solidity)
- ✓ Mobile (Kotlin & React Native)
- ✓ Web (HTML, CSS & Tailwind)

Databases

- ✓ Oracle SQL. MySQL, MongoDB, Postgres & Firebase
Proficient in Oracle SQL, MySQL, and MongoDB for database management and operations. Skilled in designing, querying, and optimizing databases for performance

Servers & Cloud Services

- ✓ Glassfish Server, Tomcat Server
I am experienced in deploying and managing applications on Glassfish and Tomcat servers. Proficient in configuring servers for optimal performance and security. I am skilled in utilizing Firebase for backend services and real-time data synchronization.
- ✓ Cloud Services
Experienced in leveraging AWS services for scalable, and secure cloud solutions. Skilled in managing resources and optimizing performance in the cloud environment.

Languages

- English
- IsiZulu
- SePedi



Projects

MedXpert Web Application

- ✓ A comprehensive health management system developed in Java using the Spring Boot framework and MySQL. MedXpert enables seamless medical record management and online appointment scheduling, enhancing patient-doctor interactions with a user-friendly interface.

ReadWell Web Application

- ✓ An advanced online book management platform built with Java (Spring Boot) and MongoDB. ReadWell allows users to read, store, and share digital books, providing an intuitive and scalable digital library experience.

Secure Document Sharing Blockchain DAPP

- ✓ A blockchain-powered decentralized application (DApp) ensuring secure and immutable document sharing. Transactions are automated via Solidity smart contracts, enhancing transparency and security. Developed using Django (Python), this system integrates blockchain technology to maintain document integrity.

Ayo-Business-Chat Mobile Application

- ✓ A business-oriented mobile communication platform where I was responsible for backend development and database design. Additionally, I developed the admin-side application in Java, integrating JavaScript, CSS, HTML, and MySQL to create a dynamic and responsive interface.

Eco-sort waste detection AI Powered Application

- ✓ An AI-driven waste detection system built with Python. This application leverages computer vision (TensorFlow) and a webcam-based detection model to analyze waste composition and guide users on proper disposal methods. Developed as part of the MEISH Hackathon, the project utilized Anaconda for virtual environment management and machine learning models to enhance accuracy.



References

Mr. K Phuduhudus

Relationship	Operating Systems, Computer Architecture & Information Security Lecture
Contact	+27 81 726 1483

Mr. S Dlamini

Relationship	Principles of Programming Lecturer
Contact	+27 79 455 6933

Mr. A Gomba

Relationship	Student Governance & Leadership Development (SGLD) Practitioner
Contact	+27 72 344 6331

Mr. D Mphogo

Relationship	Programming Mentor
Contact	+27 76 489 6252



Goals & Aspirations

- ✓ I aspire to gain valuable industry experience by working at for a technology driven organization, where I can collaborate with innovative teams and enhance my expertise in software development, data science, cybersecurity, and emerging technologies. My long-term goal is to pursue a master’s in computer science and further my education in Business Administration to develop both technical and entrepreneurial skills. With a strong passion for building secure and impactful applications, I aim to establish my own technology company, focusing on solutions that enhance security, improve efficiency, and make everyday life easier. Through continuous learning and hands-on experience, I strive to become a leader in the tech industry and drive meaningful innovation.



**UNIVERSITY OF
MPUMALANGA**

Creating Opportunities

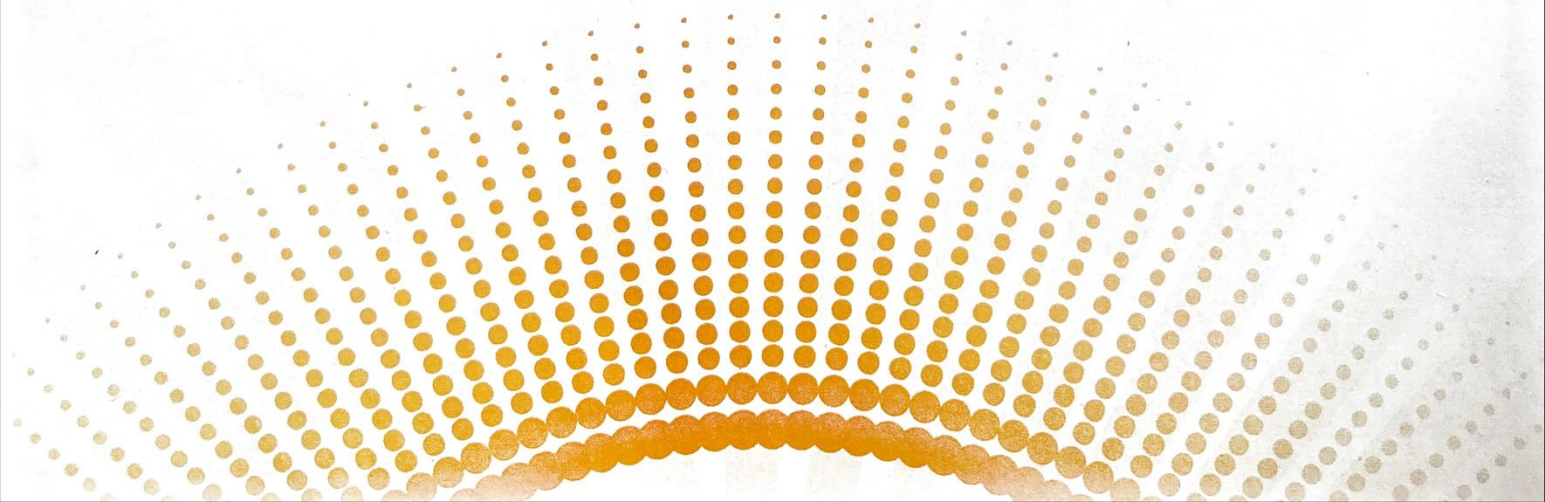
The University of Mpumalanga in collaboration with
Tshwane University of Technology, issues this certificate of attendance to

SIPHESIHLE KHUMALO

for successfully attending the
Mpumalanga Artificial Intelligence Student Hackathon

AWARDED: 2024

VICE-CHANCELLOR
Prof TV Mayekiso



CERTIFICATE OF PARTICIPATION

#ayobaBusinessChatHack

Proudly Presented To:

Siphesihle Khumalo

for having participated at the ayoba Business Chat API Hackathon (26 - 28 July 2024) and was part of a team, Binary Monks which worked on the AyoChat project.

ADJUCATED BY JUDGES FROM: 2024-07-28 12:28:33



Eero Tarjanne,
Chief Strategy Officer, ayoba

#GKHack24

GEEKULCHA />

CERTIFICATE

OF PARTICIPATION

IS PRESENTED TO

Siphesihle Khumalo

for having participated at the Geekulcha Annual Hackathon 2024 (04 - 06 October 2024) powered by Telkom and co-hosted with the Ministry for Public Service and Administration (MPSA) as a National Hackathon 2024. Siphesihle Khumalo was part of a team, Binary Monks which worked on the HomeMachine project.

ADJUCATED BY JUDGES FROM: 2024-07-28 11:03:29



MIXO NGOVENI
CEO, GEEKULCHA



the dpsa

Department:
Public Service and Administration
REPUBLIC OF SOUTH AFRICA



the cpsi

Government Component:
Centre for Public Service Innovation
REPUBLIC OF SOUTH AFRICA



Telkom

FutureMakers



**Tshwane University
of Technology**
We empower people

In accordance with the guidelines of

Directorate of Extracurricular Development

It is hereby declared that

SIPHESIHLE NQUBEKO KHUMALO

222697220

has met all the criteria for receiving a certificate for

**Top Programs Leadership Development
Most Outstanding**

for the Year

2021

Mr D Malanga
Student Development Practitioner
eMalahleni Campus

DATE



**Tshwane University
of Technology**

We empower people

In accordance with the guidelines of

Directorate of Extracurricular Development

It is hereby declared that

Siphesihle Nqubeko Khumalo

222697220

has met all the criteria for receiving a certificate for

**Top Senior Leadership Program
for the Year**

2023

Mr D Malanga
Student Development Practitioner
eMalahleni Campus

17/11/2023

DATE